ENG-0013 Incident Postmortem: INC-066

Impact

The incident involving the billing-pipeline system (System ID: SYS-010) was classified as Severity 2 (Sev2). The system, which is part of Project Echo, experienced a disruption that affected its operational stability. Given the medium criticality of billing-pipeline, the incident led to temporary delays in billing processes, impacting downstream financial reporting and customer invoicing. The resolution of the incident required 12.5 hours, during which operational capacity was compromised, necessitating immediate corrective actions to restore normal functionality.

Timeline

- **January 13, 2025**: The incident was reported by Jeffrey Chavez, a Marketing Specialist, who identified issues with the billing-pipeline system. The incident was logged under ID INC-066.
- **Immediately following report**: On-call engineer Melissa Gates, Staff Engineer for the billing-pipeline system, was alerted to investigate the issue.
- **Investigation and containment**: The root cause was identified as a deployment error, which led to the disruption.
- **Resolution**: The deployment error was rectified, and the system was restored to normal operation.
- **Total resolution time**: 12.5 hours from initial report to full recovery.

Root Cause

The root cause of the incident was determined to be a deployment error. This error originated during a recent update to the billing-pipeline system, which is built on a React/Node.js/DynamoDB tech stack. The deployment mistake was not caught during the deployment process, leading to the temporary failure of the system.

Action Items

Action Item	Responsible Party	Due Date	Notes
Implement additional deployment validation checks	DevOps Team	To be scheduled	To prevent similar deployment errors in the future
Conduct a review of recent deployment procedures	Engineering Leadership	Within one week of incident	Ensure deployment protocols are comprehensive and adhered to
Provide refresher training on deployment best practices	HR & Engineering Managers	Within two weeks	To minimize human error during deployments
	Monitoring Team	Immediate	

Action Item	Responsible Party	Due Date	Notes
Enhance monitoring and alerting for billing-pipeline			To detect issues more rapidly in future incidents
Document and update incident response procedures	Incident Response Team	Within one month	To improve response times and coordination

Additional Notes

Melissa Gates, the on-call engineer during the incident, played a critical role in diagnosing and resolving the issue swiftly. The incident was reported by Jeffrey Chavez, highlighting the importance of cross-departmental communication in incident detection. The billing-pipeline system, owned by Project Echo, remains a medium-critical component within our infrastructure, and ongoing improvements are necessary to mitigate similar risks.

This postmortem aims to ensure transparency and continuous improvement in our deployment and incident response processes.